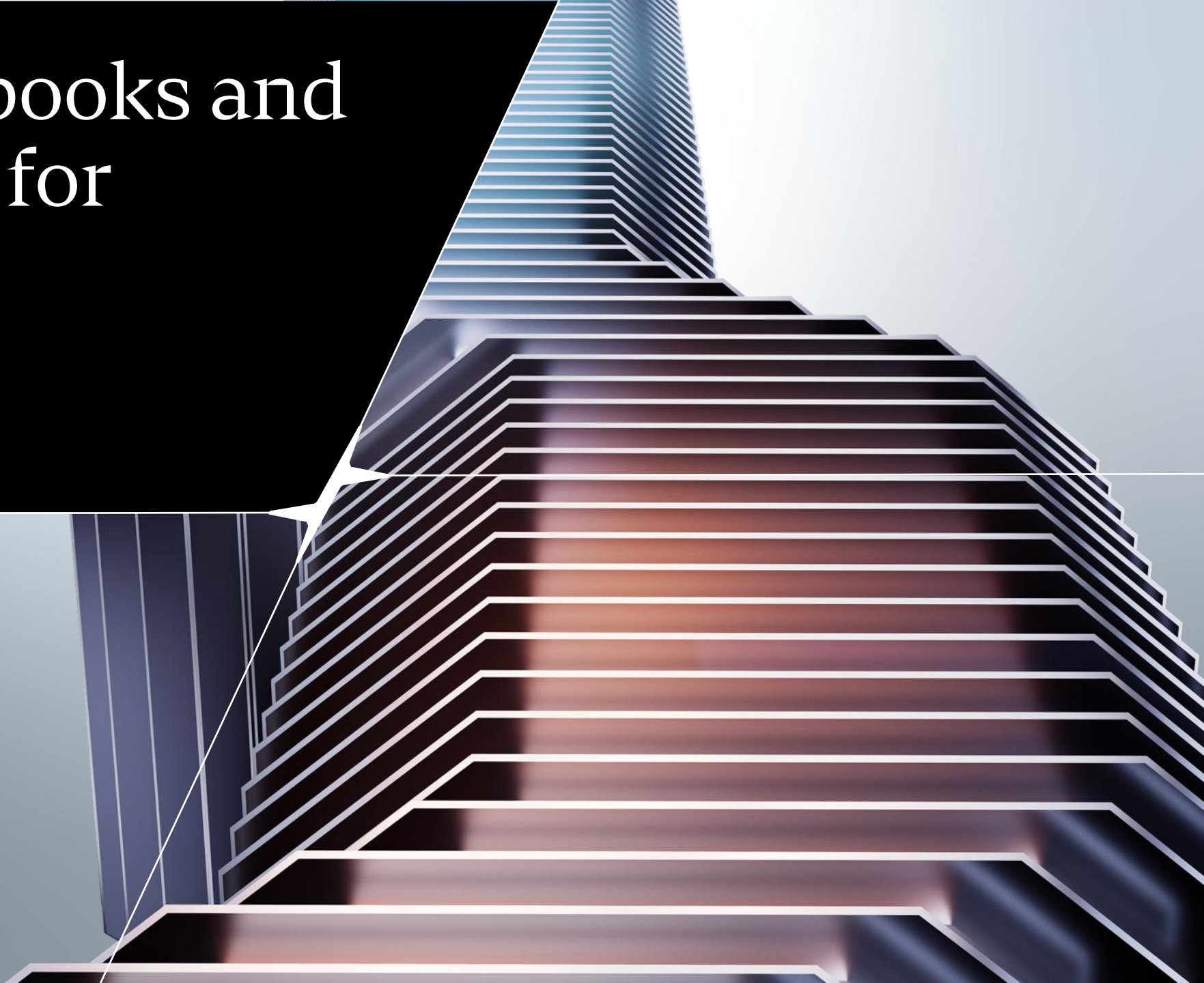


Jupyter Notebooks and Google Colab for Python/ML

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WHY NOTEBOOKS MATTER IN ML?

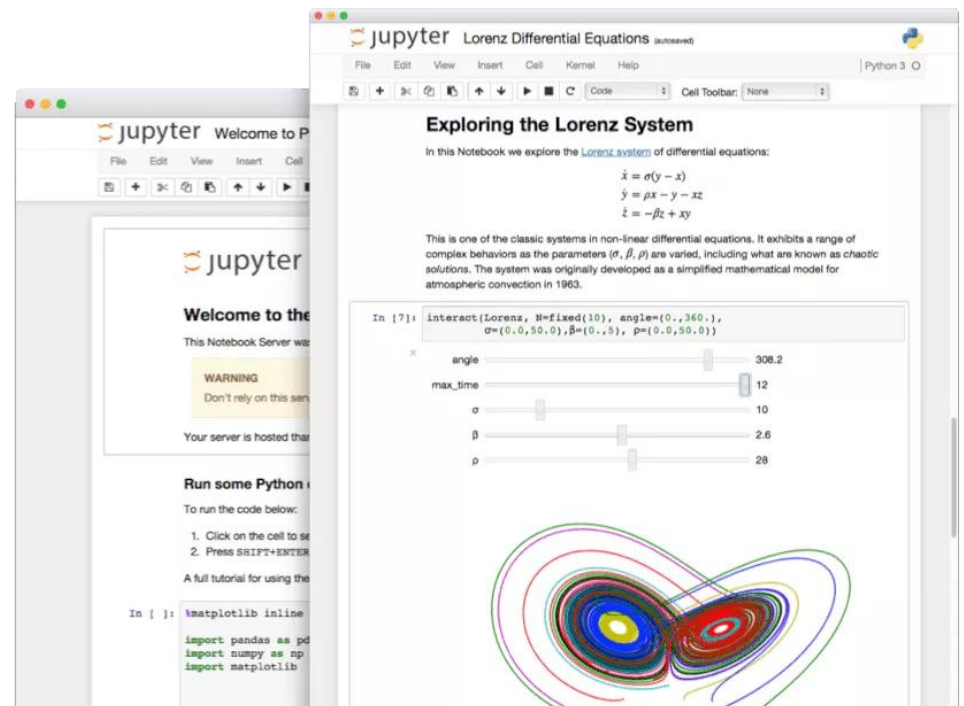
- Notebooks **combine** code, explanation, outputs, equations, charts, and experimentation in one place.
- For ML **beginners**, they are ideal for learning, exploring data, and documenting thinking.
- Notebooks are not the media to deploy models in **production** environments.





WHAT IS A JUPYTER NOTEBOOK?

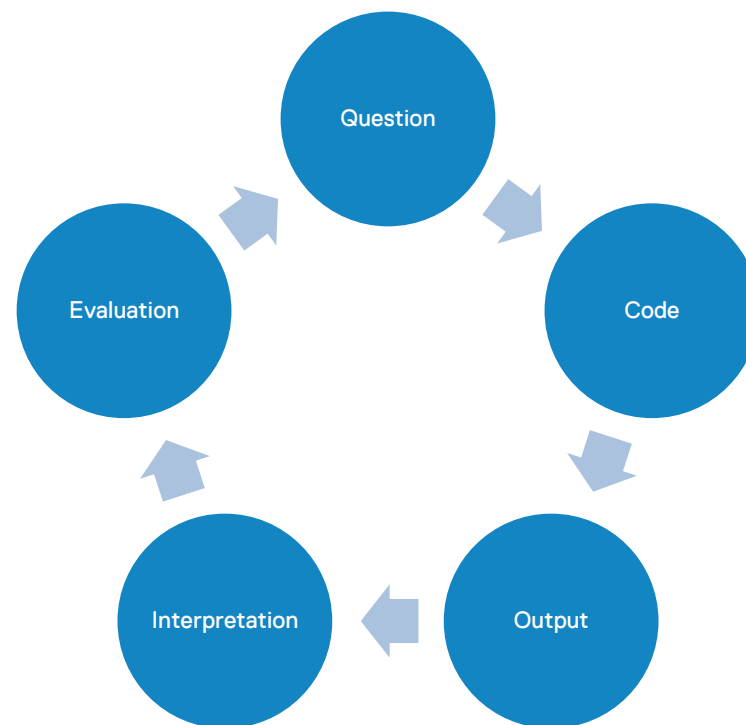
- A notebook is an **interactive document** made of cells: code cells, Markdown cells, outputs, visualizations, and rich media.
- Files usually use the **.ipynb** format.
- They have become a **standard** when running Data Science and ML experiments.





WHY DO NOTEBOOKS EXIST?

- They support **interactive** computing: write a little code, run it, inspect the result, explain what happened, and continue.
- This makes them useful for **exploration** and **communication**.
- They leverage the interactive nature of the Python **interpreter**.





ANATOMY OF A NOTEBOOK

Key parts:

- Cells
- Kernel
- Runtime
- Outputs
- Variables in memory
- File browser
- Environment/packages.

The notebook document is not the same as the running kernel.

The screenshot displays the JupyterLab user interface. On the left is a file browser showing a directory structure with files like 'bar.vl.json', 'Dockerfile', 'iris.csv', 'japan_meterolo...', 'jupiter.mp4', 'Museums_in_D...', 'README.md', 'rocket.wav', and 'zika_assembled...'. The main area is a code editor titled 'Data.ipynb' showing a Python cell with the following code:

```
[4]: import pandas
df = pandas.read_csv('../data/iris.csv')
df.head(20)
```

The output of the cell is a table with 13 rows and 4 columns: 'sepal_length', 'sepal_width', 'petal_length', and 'pet'. The data is as follows:

	sepal_length	sepal_width	petal_length	pet
0	5.1	3.5	1.4	
1	4.9	3.0	1.4	
2	4.7	3.2	1.3	
3	4.6	3.1	1.5	
4	5.0	3.6	1.4	
5	5.4	3.9	1.7	
6	4.6	3.4	1.4	
7	5.0	3.4	1.5	
8	4.4	2.9	1.4	
9	4.9	3.1	1.5	
10	5.4	3.7	1.5	
11	4.8	3.4	1.6	
12	4.8	3.0	1.4	
13	4.3	3.0	1.1	

On the right side of the interface, there is a terminal window titled 'jupyterlab.md' showing the text 'JupyterLab Demo' and 'JupyterLab: The next generation user interface for Project Jupyter'. Below this, it lists the contributors: 'Project Jupyter', 'Bloomberg', and '(then) Continuum'. At the bottom, there is a terminal window showing a file browser view of 'bar.vl.json' and a terminal window showing a file browser view of '1024px-Hubble'.



USING NOTEBOOKS OR NOT?

They are good for...

Data exploration

Visualizations

Small experiments

ML demos

Tutorials

Teaching

Reports

Reproducible walkthroughs

But, they are not that good for...

Large production systems

Complex software architecture

Hidden execution order

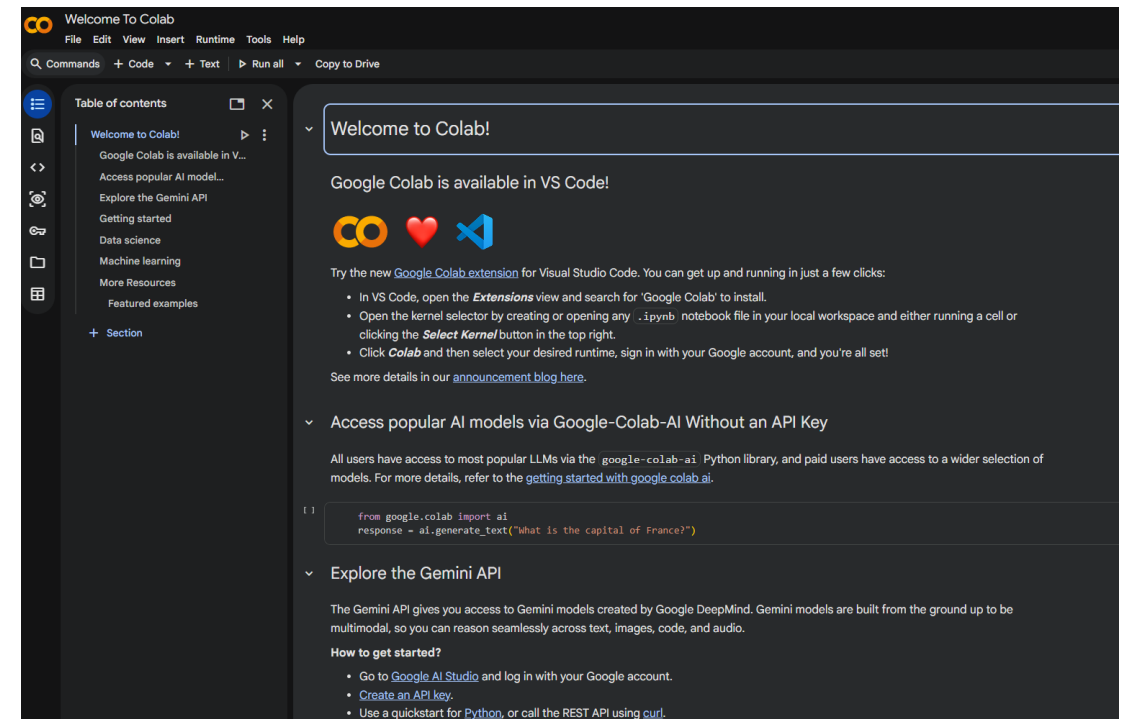
Hard-to-review long notebooks

Fragile environments



GOOGLE COLAB: NOTEBOOKS IN THE CLOUD

- Colab is a **hosted** Jupyter Notebook service.
- It requires **no local setup** and provides access to **cloud compute**, including GPUs/TPUs, with usage limits.
- Requires a **Google account** and allows persistence using Google Drive.





JUPYTER VS GOOGLE COLAB

Local Jupyter

More control

Local files

Custom environments

Works offline after setup

Give it a try here: <https://jupyter.org/try-jupyter/notebooks>

Google Colab

Fast start

Easy sharing

Cloud GPUs

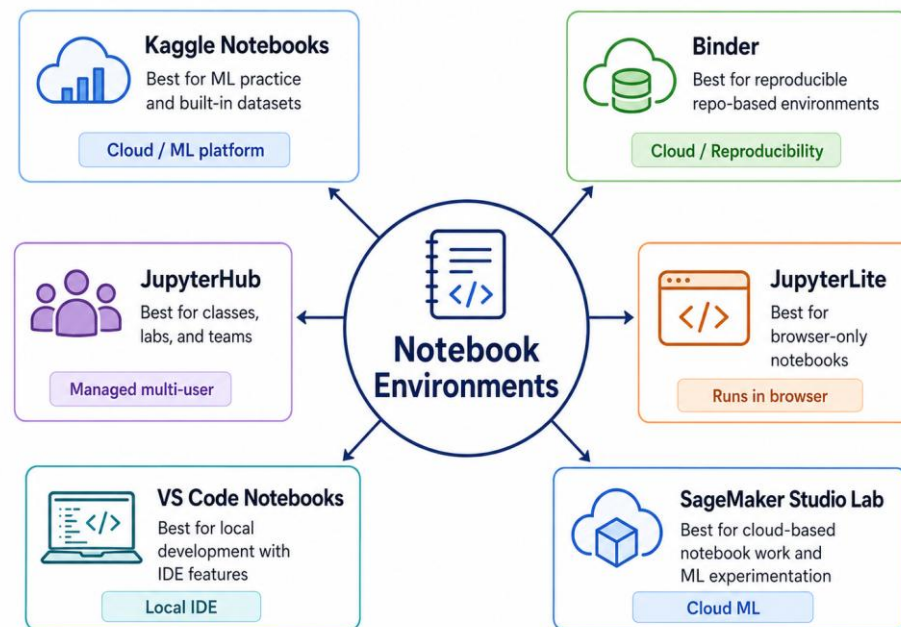
Google Drive integration

Less control and runtime/session limits

Give it a try here: <https://colab.research.google.com/notebooks/welcome.ipynb>



OTHER NOTEBOOK-BASED ENVIRONMENTS



☆ Different environments, different strengths.



SOME USEFUL RESOURCES

- Jupyter Project: <https://jupyter.org/>
 - Jupyter Notebooks: <https://docs.jupyter.org/en/latest>
 - Notebook Narratives: <https://docs.jupyter.org/en/stable/use/use-cases/narrative-notebook.html>
 - Installing Jupyter: <https://jupyter.org/install>
 - Google Colab: <https://colab.research.google.com/>
 - Kaggle Notebooks: <https://www.kaggle.com/docs/notebooks>
 - Jupyter Binder: <https://jupyter.org/binder>
 - Jupyter Notebooks in VS Code: <https://code.visualstudio.com/docs/datascience/jupyter-notebooks>
 - JupyterLite: <https://jupyterlite.readthedocs.io/en/stable>
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Thanks!

Any further questions?

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